

CD — Colour by Design

OCR B (Salters) H433 storyline **CD**. Dyes and colour to order: the origins of colour in delocalised systems, arenes and their reactions, diazonium compounds and dyeing, fats and oils, gas-liquid chromatography and carbonyl chemistry.

1. Aromatic Chemistry

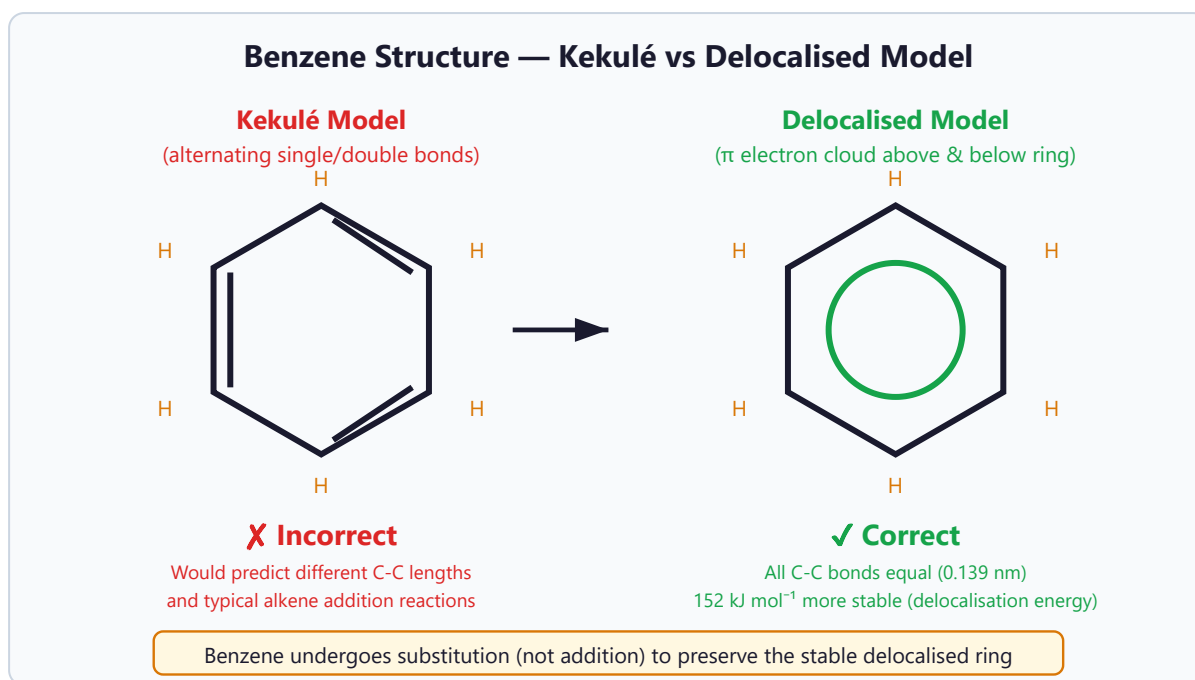
Structure of Benzene

Benzene (C_6H_6): A planar, cyclic hydrocarbon with a ring of six carbon atoms. Each carbon is sp^2 hybridised with bond angles of 120° . The unhybridised p orbital on each carbon overlaps sideways with those on adjacent carbons, forming a delocalised π system above and below the plane of the ring.

Kekulé model: Originally proposed as alternating single and double bonds (cyclohexatriene). However, this model is inconsistent with experimental evidence:

- All C–C bond lengths in benzene are equal (0.139 nm), intermediate between a single bond (0.154 nm) and a double bond (0.134 nm).
- Benzene does not undergo addition reactions typical of alkenes (e.g. it does not readily decolourise bromine water).
- The enthalpy of hydrogenation of benzene (-208 kJ mol^{-1}) is less exothermic than predicted for three C=C bonds ($3 \times -120 = -360 \text{ kJ mol}^{-1}$), showing benzene is more stable than the hypothetical cyclohexatriene by about 152 kJ mol^{-1} (the delocalisation/resonance energy).

Delocalised model: The six π electrons are spread evenly over all six carbon atoms in a ring of electron density. This delocalisation makes benzene more thermodynamically stable than the Kekulé structure and explains its tendency to undergo substitution rather than addition reactions (preserving the stable delocalised ring).



Electrophilic Substitution of Benzene

The delocalised π electron ring is a region of high electron density that attracts electrophiles. However, benzene undergoes substitution rather than addition to preserve the stability of the aromatic ring.

Nitration:



The electrophile is the nitronium ion (NO_2^+), generated by the reaction of concentrated nitric acid with concentrated sulfuric acid: $\text{HNO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{NO}_2^+ + \text{HSO}_4^- + \text{H}_2\text{O}$. The mechanism involves the NO_2^+ being attracted to the π system, forming a positively charged intermediate (Wheland intermediate), followed by loss of H^+ to restore aromaticity. Temperature must be kept below 55°C to prevent dinitration.

Halogenation:



A halogen carrier catalyst (Lewis acid such as AlBr_3 , FeBr_3 , or AlCl_3) is needed to generate the electrophile. The catalyst polarises the $\text{Br}-\text{Br}$ bond or forms Br^+ : $\text{Br}_2 + \text{AlBr}_3 \rightarrow \text{Br}^+ + \text{AlBr}_4^-$. The Br^+ then attacks the π ring.

Sulfonation:



Benzene is refluxed with fuming sulfuric acid (concentrated sulfuric acid with dissolved sulfur trioxide) to give benzenesulfonic acid. The electrophile is the sulfur trioxide molecule itself, SO_3 : the sulfur atom carries a substantial δ^+ charge because it is bonded to three electronegative oxygen atoms, so it is attacked by the π ring in the usual two steps — the ring donates a pair of electrons to sulfur, and the intermediate then loses H^+ to restore aromaticity. With ordinary concentrated sulfuric acid alone the reaction is very slow, which is why the fuming acid (or several hours of reflux) is used.

Sulfonation matters in this storyline because of what the $-\text{SO}_3\text{H}$ group does to a molecule. It is a strong acid, so in water it exists as the sulfonate ion $-\text{SO}_3^-$. Attaching one or more sulfonate groups to a large, flat, otherwise insoluble dye molecule makes it soluble enough to be applied from an aqueous dyebath, and gives it a negative charge that can form an ionic bond to the protonated $-\text{NH}_3^+$ groups of wool, silk and nylon. Almost every acid dye you meet in CD carries sulfonate groups for exactly these two reasons.

Friedel-Crafts alkylation:



The AlCl_3 catalyst generates a carbocation electrophile (R^+) from the halogenoalkane. This introduces an alkyl group onto the benzene ring.

Friedel-Crafts acylation:



The electrophile is an acylium ion (RCO^+). This introduces an acyl group ($\text{C}=\text{O}$ bonded to R) onto the ring. Acylation is preferred over alkylation in synthesis because it does not give multiple substitutions (the $\text{C}=\text{O}$ group deactivates the ring).

Exam Tip: In all electrophilic substitution mechanisms on benzene, draw two curly arrows: one from the π ring to the electrophile (forming the new bond), and one from the $\text{C}-\text{H}$ bond to the ring (restoring aromaticity by losing H^+). Show the intermediate Wheland complex with its positive charge delocalised over the ring.

2. Carbonyl Compounds

Aldehydes and Ketones

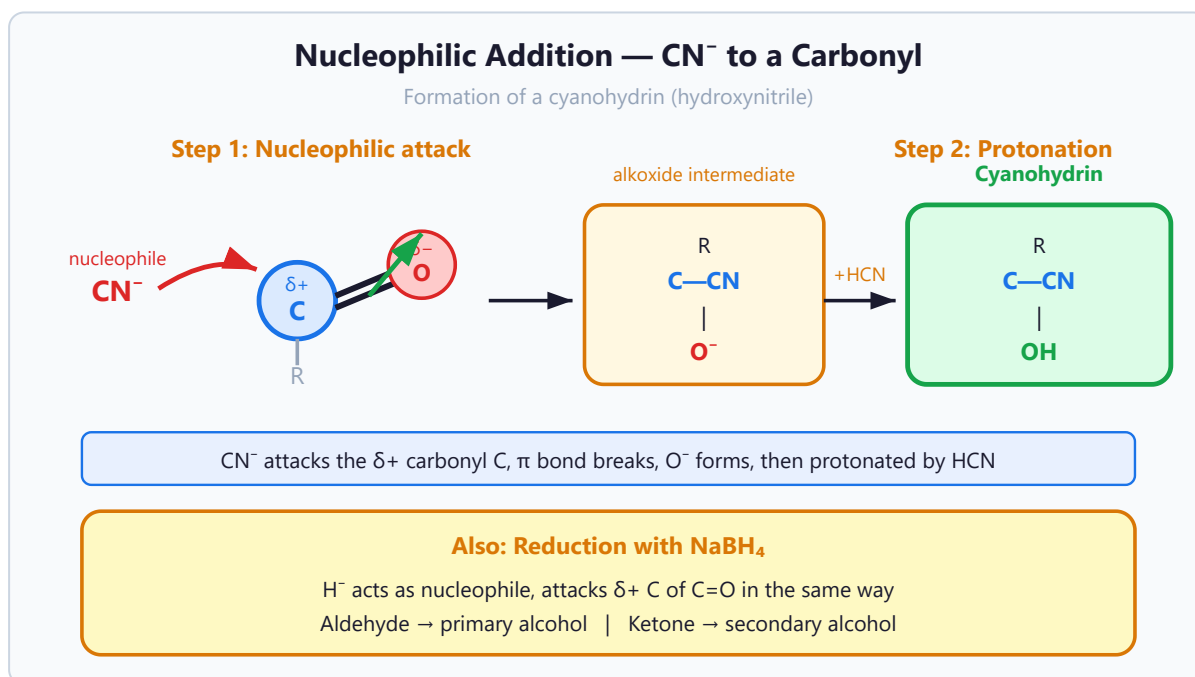
Both contain the C=O (carbonyl) group. Aldehydes have the carbonyl at the end of the chain (RCHO); ketones have it within the chain (RCOR'). The C=O bond is polar ($\delta+$ on C, $\delta-$ on O) because oxygen is more electronegative than carbon, making the carbonyl carbon susceptible to nucleophilic attack.

Nucleophilic Addition

The key mechanism for carbonyl compounds. A nucleophile attacks the $\delta+$ carbon of the C=O group:

With HCN (hydrogen cyanide): The nucleophile is CN^- . It attacks the $\delta+$ carbon, breaking the π bond and forming an alkoxide intermediate, which is then protonated by HCN to give a hydroxynitrile (cyanohydrin). This reaction extends the carbon chain by one carbon atom and introduces both an OH and CN group. A trace of base (NaCN or KCN) is used as a catalyst to generate CN^- .

With NaBH_4 (sodium borohydride): A reducing agent. The H^- ion acts as the nucleophile, attacking the $\delta+$ carbonyl carbon. Aldehydes are reduced to primary alcohols; ketones are reduced to secondary alcohols. NaBH_4 is used in aqueous/ethanolic solution.



Distinguishing Aldehydes from Ketones

Test	Reagent	Aldehyde result	Ketone result
Tollens' test	$[\text{Ag}(\text{NH}_3)_2]^+$ (silver mirror reagent), warm	Silver mirror forms (Ag^+ reduced to Ag)	No reaction
Fehling's test	Fehling's solution (Cu^{2+} tartrate complex), heat	Brick-red precipitate of Cu_2O	No reaction
2,4-DNP test	2,4-dinitrophenylhydrazine (Brady's reagent)	Orange/yellow precipitate (confirms $\text{C}=\text{O}$ present)	Orange/yellow precipitate (confirms $\text{C}=\text{O}$ present)

Tollens' and Fehling's tests work because aldehydes can be oxidised to carboxylic acids, but ketones cannot be oxidised under these mild conditions. The 2,4-DNP test confirms the presence of a carbonyl group (aldehyde or ketone) but does not distinguish between them. The melting point of the 2,4-DNP derivative can be used to identify the specific aldehyde or ketone.

3. Colour, Dyes and Dyeing

Where Colour Comes From

A molecule is coloured because it absorbs some wavelengths of visible light and not others. Absorbing a photon promotes an electron from a filled to an empty energy level, and the photon absorbed is the one whose energy exactly matches the gap.

Chromophore: The part of a molecule responsible for absorbing light — in organic dyes, a system of delocalised electrons. Typical chromophores are conjugated chains of alternating double and single bonds, aromatic rings and the azo group, $-\text{N}=\text{N}-$.

The key relationship is between the *extent* of the delocalisation and the *energy* absorbed:

- A short delocalised system has widely spaced energy levels, so a large energy gap, so it absorbs short-wavelength, high-energy radiation — in the ultraviolet. The compound looks colourless.
- Extending the delocalisation brings the levels closer together, so the gap shrinks, so the absorption moves to longer wavelengths — into the visible. The compound is coloured.
- Extending it further moves the absorption further towards the red end.

Benzene, with delocalisation over one ring, absorbs in the UV and is colourless. Beta-carotene, the pigment of carrots, has eleven conjugated C=C bonds; its delocalised system is long enough that it absorbs blue-violet light, and it looks orange.

Complementary colour: What we see is the light that is *not* absorbed. A substance that absorbs one colour appears the complementary colour.

Colour absorbed	Approximate wavelength absorbed / nm	Colour seen
Violet	400–430	Yellow
Blue	430–490	Orange
Green	490–560	Red
Yellow	560–590	Violet
Orange	590–630	Blue
Red	630–700	Green

Diazonium Compounds

Azo dyes — the largest single class of synthetic dye — are made in two steps, and both conditions matter.

Step 1: diazotisation. Phenylamine is treated with nitrous acid, made *in situ* from sodium nitrite and hydrochloric acid, at a temperature below 10 °C:

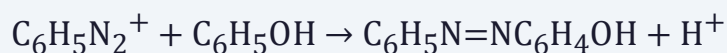


The product is benzenediazonium chloride. The low temperature is essential: above about 10 °C the diazonium ion decomposes, losing nitrogen gas and giving phenol instead, which ruins the preparation.



Step 2: coupling. The cold diazonium solution is added to an alkaline solution of a phenol or an aromatic amine. The diazonium ion acts as an electrophile and substitutes into the other ring, usually at the position

opposite the -OH or -NH_2 group:



The product is an azo compound, and it is strongly coloured. The -N=N- group joins the two rings into one continuous delocalised system spanning the whole molecule, which is why an azo dye absorbs in the visible when neither starting material does. Alkaline conditions are used for the coupling because they convert the phenol to the phenoxide ion, which is more electron-rich and so reacts faster with the electrophile.

Different coupling partners give different colours: phenol gives yellow-orange dyes, naphthalen-2-ol gives reds, and aromatic amines give oranges and browns. Historically, Perkin's mauveine of 1856 was the first commercially successful synthetic dye, and picric acid and chrysoidine were among the early industrial products that made the chemistry of colour worth money.

The Anatomy of a Dye Molecule

A commercial dye is designed, not stumbled upon, and different parts of the molecule do different jobs:

Part of the molecule	Job	Typical groups
Chromophore	Absorbs visible light — gives the dye its colour	-N=N- , conjugated C=C chains, aromatic rings, C=O in conjugation
Groups that modify the chromophore	Shift the exact shade by donating or withdrawing electron density into the delocalised system	-OH , -NH_2 , -NR_2 , -NO_2
Groups that affect solubility	Make the dye dissolve in the dye bath, or deliberately keep it insoluble once fixed	$\text{-SO}_3^-\text{Na}^+$, $\text{-COO}^-\text{Na}^+$, -OH
Groups that bond to the fibre	Hold the dye on the cloth through washing	-SO_3^- and -NH_3^+ for ionic bonding; reactive triazine rings for covalent bonding

How Dyes Attach to Fibres

"Fastness" — how well a dye survives washing and light — depends on the kind of bonding between dye and fibre, and the order is what you would expect from bond strengths.

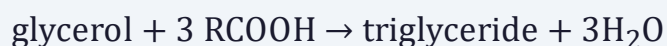
Type of dye	Bonding to the fibre	Example / fibre	Fastness
Direct dye	Intermolecular bonds only — instantaneous dipole-induced dipole bonds and hydrogen bonds. Long, flat molecules lie along the fibre and make many contacts	Cotton (cellulose), which is rich in -OH groups	Lowest — washes out over time
Acid and basic dyes	Ionic bonds between a charged group on the dye and an oppositely charged group on the fibre	Wool, silk and nylon, whose chains carry -NH_3^+ and -COO^- groups	Good
Mordant dye	A metal ion such as Al^{3+} or Cr^{3+} is fixed on the fibre first and then bonds to the dye, bridging the two	Traditional wool dyeing with alum as the mordant	Good; the mordant also changes the shade
Fibre-reactive dye	Covalent bonds formed between the dye and -OH groups of the cellulose	Cotton dyed with triazine-based reactive dyes	Highest — the dye is part of the fibre
Vat dye	The insoluble dye is reduced to a soluble form, soaks into the fibre, and is then re-oxidised in air so that the insoluble dye is trapped inside	Indigo on denim	Good, though it rubs off the surface — hence faded jeans

Exam Tip: Questions on dyeing usually give you a dye structure and ask you to identify which group does what. Work through it systematically: find the extended delocalised system (the chromophore), then look for ionic or polar groups such as -SO_3^- (solubility, and ionic bonding to wool or nylon), and then for anything that could form a covalent bond to an -OH group of cellulose. You are not expected to recall dye structures — they are given — only to interpret them.

4. Fats and Oils

Fats and oils: Naturally occurring mixed esters of propane-1,2,3-triol (glycerol) with long-chain carboxylic acids (fatty acids). Because all three –OH groups of the glycerol are esterified, they are also called triglycerides or triesters.

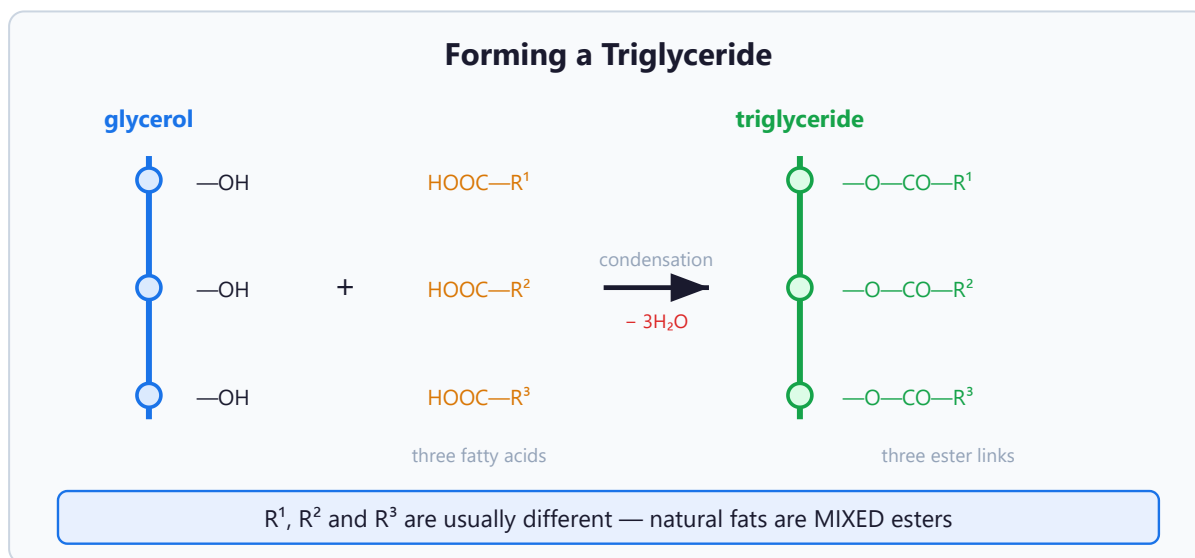
A triglyceride is formed by three separate esterification (condensation) reactions, one at each –OH group, with a molecule of water lost each time:



Taking octadecanoic (stearic) acid, $\text{C}_{17}\text{H}_{35}\text{COOH}$, as the fatty acid:



Check it balances by mass: $92 + (3 \times 284) = 944$ on the left, and $890 + (3 \times 18) = 944$ on the right. "Mixed" ester means the three R groups in a natural fat are usually *not* all the same.



Saturated and Unsaturated

	Saturated	Unsaturated
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Fatty acid chains	No C=C double bonds	One (mono-) or more (poly-) C=C double bonds, normally <i>cis</i>
Shape of the chain	Straight and flexible	Kinked at each <i>cis</i> double bond
Packing	Chains pack closely and regularly	Kinks stop the chains packing closely
Intermolecular bonds	More contact, so stronger instantaneous dipole-induced dipole bonds	Less contact, so weaker bonds
Melting point	Higher — a solid fat at room temperature	Lower — a liquid oil at room temperature
Example acid	Stearic acid, C ₁₇ H ₃₅ COOH (M _r 284)	Oleic acid, C ₁₇ H ₃₃ COOH (M _r 282), one C=C

The whole difference between a fat and an oil is therefore two hydrogen atoms and a kink. Note that the argument is about *shape and packing*, not about the strength of the C=C bond itself — a very common confusion.

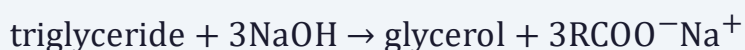
Reactions of Fats and Oils

Testing for unsaturation. The C=C bonds decolourise bromine water by electrophilic addition. The volume of bromine or iodine that a known mass of fat absorbs measures how unsaturated it is.

Hydrogenation. Passing hydrogen through a vegetable oil over a nickel catalyst at about 150–200 °C adds H₂ across some of the C=C bonds. The chains become straighter, pack better, and the melting point rises, turning a liquid oil into a spreadable solid — this is how margarine is made.



Hydrolysis. Being esters, fats are hydrolysed back to glycerol and fatty acids. With hot aqueous acid the products are glycerol and the free carboxylic acids; with hot aqueous alkali, the process called **saponification**, the products are glycerol and the sodium salts of the fatty acids — soap.



Soap cleans because each anion has a long non-polar hydrocarbon tail that mixes with grease and an ionic -COO^- head that is attracted to water.

Worked Example — Saponification:

Calculate the mass of glycerol obtained by completely hydrolysing 89.0 g of tristearin, $\text{C}_{57}\text{H}_{110}\text{O}_6$ ($M_r = 890$), with sodium hydroxide. (M_r glycerol $\text{C}_3\text{H}_8\text{O}_3 = 92.0$)

$$n(\text{tristearin}) = 89.0 / 890 = 0.100 \text{ mol}$$

Each triglyceride gives one glycerol, so $n(\text{glycerol}) = 0.100 \text{ mol}$

$$\text{Mass of glycerol} = 0.100 \times 92.0 = \mathbf{9.20 \text{ g}}$$

The soap produced is 0.300 mol of sodium stearate, $\text{C}_{17}\text{H}_{35}\text{COO}^-\text{Na}^+$ ($M_r = 306$), a mass of 91.8 g.

Check: 89.0 g of fat plus $0.300 \times 40.0 = 12.0 \text{ g}$ of NaOH gives 101.0 g of products, and $9.20 + 91.8 = 101.0 \text{ g}$. Consistent.

Analysing Fats by Gas-Liquid Chromatography

The food-testing context of this storyline needs a way of finding out which fatty acids a fat actually contains. The fat is hydrolysed, the fatty acids are converted to their volatile methyl esters, and the mixture is injected into a gas-liquid chromatograph. The sample is vaporised and carried by an inert gas through a column holding a high-boiling liquid on a porous support; each ester spends a different proportion of its time dissolved in the stationary phase, so each emerges at a different **retention time**. Comparing retention times with standards identifies each fatty acid, and peak areas give the proportions. Feeding the outflow into a mass spectrometer (GC-MS) identifies components that no standard is available for.

Exam Tip: If asked why an unsaturated oil has a lower melting point than a saturated fat, three things must appear for full marks: the *cis* $\text{C}=\text{C}$ introduces a kink; the kinked chains cannot pack as closely; so the instantaneous dipole-induced dipole bonds between molecules are weaker and less energy is needed to separate them. Saying only "it has double bonds" earns nothing.